

习题 14.2

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提示

$$\lim_{n \rightarrow \infty} \frac{a^n}{n!} = 0$$

第二章, §1 例 6.

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- (1)

由例 3 可知

$$e^x = 1 + \frac{1}{1!}x + \frac{1}{2!}x^2 + \cdots + \frac{1}{n!}x^n + \cdots, x \in (-\infty, \infty)$$

故

$$e^{x^2} = 1 + \frac{1}{1!}x^2 + \frac{1}{2!}(x^2)^2 + \cdots + \frac{1}{n!}(x^2)^n + \cdots, x \in (-\infty, \infty)$$

- (2)

由例 6 可知 ($\alpha \leq -1$) 时,

$$(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \cdots + \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!}x^n + \cdots, x \in (-1, 1)$$

$\alpha = -1, x = -x$ 得到

$$\begin{aligned} \frac{x^{10}}{1-x} &= x^{10} \left(1 + (-1)(-x) + \frac{(-1)(-2)}{2}(-x)^2 + \cdots + \frac{(-1)(-2)\cdots(-1-n+1)}{n!}(-x)^n + \cdots \right) \\ &= x^{10} (1 + x + x^2 + \cdots + x^n + \cdots), x \in (-1, 1) \end{aligned}$$

• (3)

由例 6 可知 $(-1 < \alpha < 0)$ 时,

$$(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \cdots + \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!}x^n + \cdots, x \in (-1, 1]$$

令 $x = -2x, \alpha = -\frac{1}{2}$ 时得到

$$\begin{aligned} \frac{1}{\sqrt{1-2x}} &= 1 + \left(-\frac{1}{2}\right)(-2x) + \frac{\left(-\frac{1}{2}\right)\left(-\frac{1}{2}-1\right)}{2!}(-2x)^2 \\ &\quad + \cdots + \frac{-\frac{1}{2}\left(-\frac{1}{2}-1\right)\cdots\left(-\frac{1}{2}-n+1\right)}{n!}(-2x)^n + \cdots \\ &= 1 + \left(\frac{1}{2}\right)(2x) + \frac{\left(\frac{1}{2}\right)\left(\frac{1}{2}+1\right)}{2!}(2x)^2 \cdots + \frac{\frac{1}{2}\left(\frac{1}{2}+1\right)\cdots\left(\frac{1}{2}+n-1\right)}{n!}(2x)^n + \cdots \\ &= 1 + x + \frac{\left(\frac{1\cdot 3}{2^2}\right)}{2!}(2x)^2 \cdots + \frac{\frac{1\cdot 3\cdot 5\cdots(1+2n-2)}{2^n}}{n!}(2x)^n + \cdots \\ &= 1 + x + \frac{1\cdot 3}{2!}x^2 \cdots + \frac{1\cdot 3\cdot 5\cdots(2n-1)}{n!}x^n + \cdots \\ &= 1 + x + \frac{1\cdot 3}{2!}x^2 \cdots + \frac{(2n-1)!!}{n!}x^n + \cdots \end{aligned}$$

故

$$\begin{aligned} \frac{x}{\sqrt{1-2x}} &= x \left(1 + x + \frac{1\cdot 3}{2!}x^2 \cdots + \frac{(2n-1)!!}{n!}x^n + \cdots \right) \\ &= x + \sum_{n=1}^{\infty} \frac{(2n-1)!!}{n!}x^{n+1} \end{aligned}$$

且收敛区域为

$$\begin{aligned} -1 &< -2x \leq 1 \\ -\frac{1}{2} &\leq x < \frac{1}{2} \end{aligned}$$